electrolytic solution was measured as 15 Ω . Find the specific conductance of the solution.

Solution

$$\kappa = \frac{1}{R}\left(\frac{l}{A}\right)$$

$$\kappa = \frac{1}{15\Omega} \times \frac{1.5\times10^{-2}\,m}{4.5\times10^{-4}\,m^2}$$

$$= 2.22\ Sm^{-1}$$

$l = 1.5\ cm = 1.5\times10^{-2}m$

$A = 4.5\ cm^2 = 4.5\times(10^{-4})m^2$

$R = 15\Omega$

9.1.1 Molar conductivity (Λ_m)

Solutions of different concentrations have different number of electrolytic ions in a given volume of solution and hence they have different specific conductance. Therefore a new quantity called molar conductance (Λ_m) was introduced.

Let us imagine a conductivity cell in which the electrodes are separated by $1m$ and having $V\ m^3$ of electrolytic solution which contains 1 mole of electrolyte. The conductance of such a system is called the molar conductance (Λ_m)

We have just learnt that the conductance of $1\ m^3$ electrolytic solution is called the specific conductance (κ). Therefore, the conductance of the above mentioned $V\ m^3$ solution (Λ_m) is given by the following expression.

$$(\Lambda_m) = \kappa \times V \qquad .....(9.6)$$

We know that, molarity (M) = $\frac{\text{Number of moles of solute (n)}}{\text{Volume of the solution (V in dm}^3\text{)}}$

Therefore, Volume of the solution containing one mole of solute = $\frac{1}{M}\ \left(mol^{-1}\ L\right)$

$$\therefore \text{Volume per } m^3\ (V) = \frac{10^{-3}}{M}\left(mol^{-1}m^3\right)$$

Substitute (9.7) in (9.6)

$$(9.6) \Rightarrow \Lambda_m = \frac{\kappa\ (Sm^{-1}) \times 10^{-3}}{M}\ mol^{-1}m^{-3} \qquad .....(9.8)$$

The above relation defines the molar conductance in terms of the specific conductance and the concentration of the electrolyte.

Example

Calculate the molar conductance of 0.025M aqueous solution of calcium chloride at $25^{\circ}C$. The specific conductance of calcium chloride is $12.04 \times 10^{-2}\ Sm^{-1}$.

$$\text{Molar conductance} = \Lambda_m = \frac{\kappa\ (Sm^{-1}) \times 10^{-3}}{M}\ mol^{-1}m^3$$

$$= \frac{(12.04 \times 10^{-2}\ Sm^{-1}) \times 10^{-3}\left(mol^{-1}m^3\right)}{0.025}$$

$$= 481.6\times10^{-5}\ Sm^2mol^{-1}$$

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